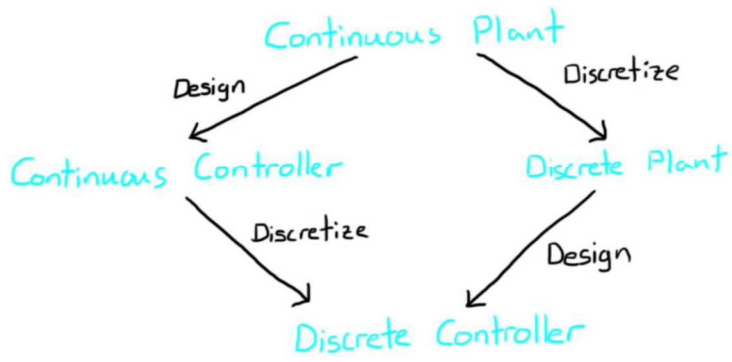
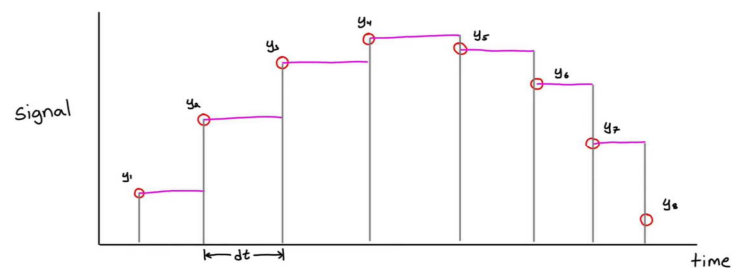
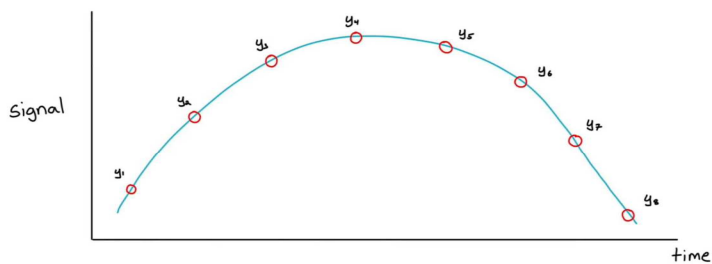
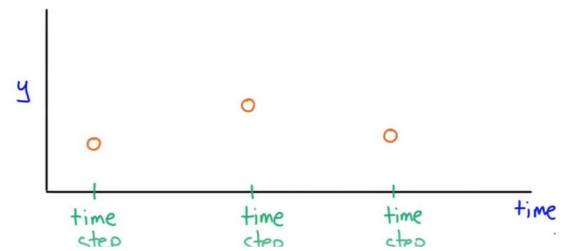
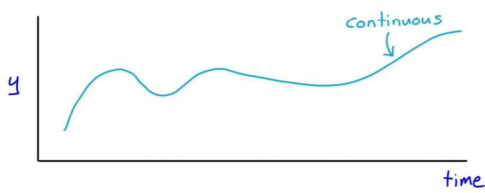
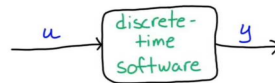


$$\frac{Y(s)}{U(s)} = \frac{s+2}{s^2+s+2}$$

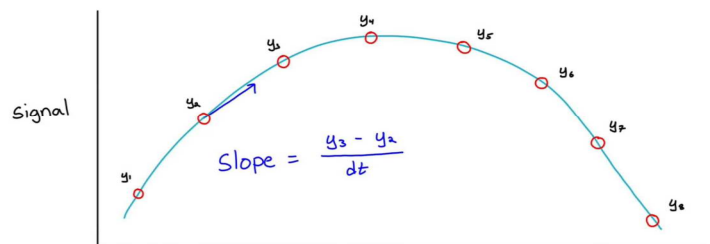
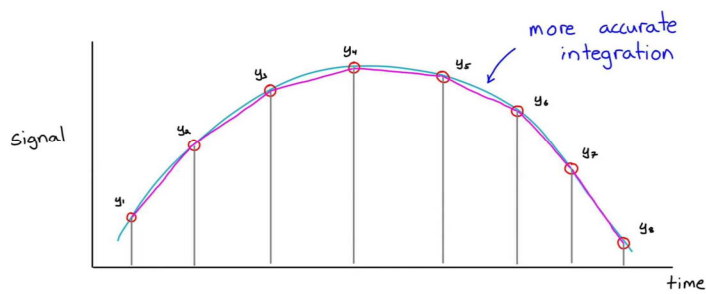
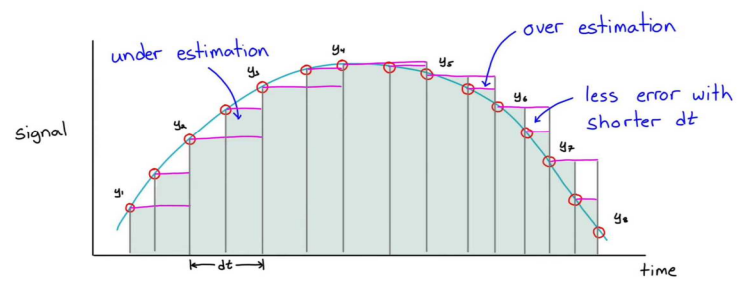
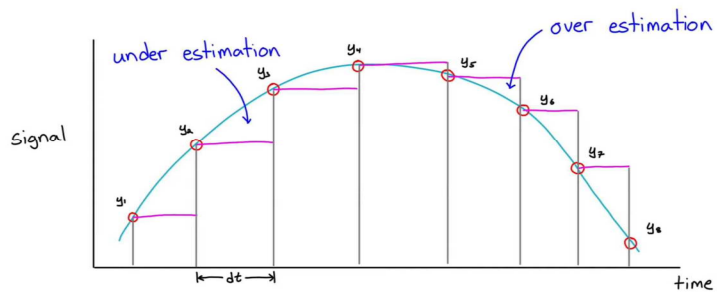


```

for i = 1:length(t)
  if i == 1
    ud(i) = ud0;
    yd(i) = yd0;
    y(i) = y0;
    ...
  end
end
  
```



$$\text{area} = y_1 \cdot dt + y_2 \cdot dt + \dots$$



1. Numerically solve the differential equation

$$\frac{Y(s)}{U(s)} = \frac{s + 2}{s^2 + s + 2}$$

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$$\ddot{y}(t) + \dot{y}(t) + 2y(t) = \dot{u}(t) + 2u(t)$$

first time step

$$y_{dd} = u_d + 2u - y_d - y$$

Second time step

$$y_d = \text{integral}(y_{dd})$$

$$y = \text{integral}(y_d)$$

$$u = \text{input}$$

$$u_d = \text{derivative}(u)$$

$$y_{dd} = u_d + 2u - y_d - y$$

1. Numerically solve the differential equation

$$\frac{Y(s)}{U(s)} = \frac{s+2}{s^2+s+2}$$

$$(s^2 + s + 2) Y(s) = (s + 2) U(s)$$

$$\ddot{y}(t) + \dot{y}(t) + 2y(t) = \dot{u}(t) + 2u(t)$$

$$\ddot{y}(t) = \dot{u}(t) + 2u(t) - \dot{y}(t) - 2y(t)$$

2. Convert and discretize in a state space

s-domain transfer function $\Rightarrow$ time-domain state space
 $\text{sys} = \text{ss}(G)$

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s-domain transfer function $\Rightarrow$ time-domain state space
 $\text{sys} = \text{ss}(G)$

$$\dot{X}(t) = A X(t) + B u(t)$$

$$y(t) = C X(t) + D u(t)$$

first time step

$$X_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$u = \text{input}$$

$$X_d = A X_0 + B u$$

$$y = C X_0 + D u$$

Second time step

$$X = \text{integral}(X_d)$$

$$u = \text{input}$$

$$X_d = A X + B u$$

$$y = C X + D u$$

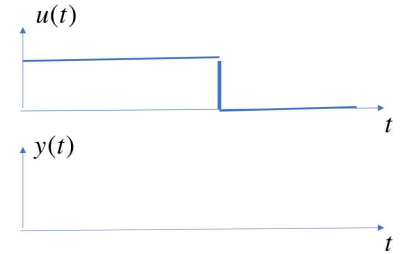
3. Convert to z-domine

$$G(s) = \frac{Y(s)}{U(s)} \Rightarrow H(z^{-1}) = \frac{Y(z^{-1})}{U(z^{-1})}$$

$$G(s) = \frac{Y(s)}{U(s)} \Rightarrow H(z^{-1}) = \frac{Y(z^{-1})}{U(z^{-1})}$$

$$s = \frac{1}{\tau} (1 - z^{-1})$$

$$G(s) = \frac{1}{Ts}$$



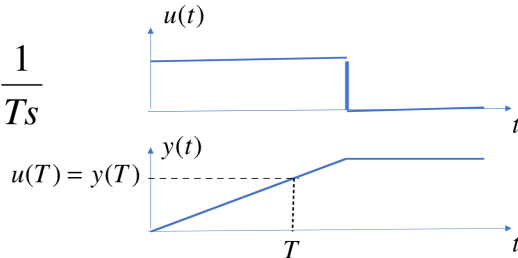
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$$G(s) = \frac{1}{Ts} \Rightarrow H(z^{-1}) = \frac{1}{T \cdot \frac{1}{\tau} (1 - z^{-1})} = \frac{Y(z^{-1})}{U(z^{-1})}$$

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$$Y \cdot T \cdot \frac{1}{\tau} (1 - z^{-1}) = U$$

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$$Y - Y \cdot z^{-1} = U \cdot \tau \cdot \frac{1}{T}$$

$$H(z^{-1}) = \frac{1}{T \cdot \frac{1}{\tau} (1 - z^{-1})} = \frac{Y(z^{-1})}{U(z^{-1})}$$

$$Y \cdot T \cdot \frac{1}{\tau} (1 - z^{-1}) = U$$

$$Y - Y \cdot z^{-1} = U \cdot \tau \cdot \frac{1}{T}$$

$$y(n) - y(n-1) = u(n) \cdot \tau \cdot \frac{1}{T}$$

$$y(n) - y(n-1) = u(n) \cdot \tau \cdot \frac{1}{T}$$

$$y(n) = y(n-1) + u(n) \cdot \tau \cdot \frac{1}{T}$$

$$\tau \ll T$$

$$\tau \leq \frac{1}{5} T$$

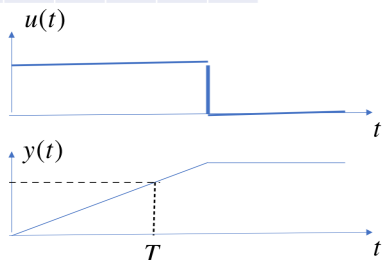
$$\tau = 0.2 \text{ s}, \quad T = 1 \text{ s}$$

n	1	2	3	4	5	6	7	8	9
u(n)	1	1	1	1	1	1	1	0	0
y(n)	0								

$$y(n) = y(n-1) + u(n) \cdot \tau \cdot \frac{1}{T}$$

$$\tau = 0.2 \text{ s}, \quad T = 1 \text{ s}$$

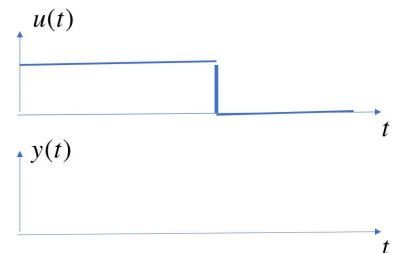
n	1	2	3	4	5	6	7	8	9
u(n)	1	1	1	1	1	1	1	0	0
y(n)	0	0.2	0.4	0.6	0.8	1.0	1.2	1.2	1.2



$$G(s) = \frac{Y(s)}{U(s)} \Rightarrow H(z^{-1}) = \frac{Y(z^{-1})}{U(z^{-1})}$$

$$s = \frac{1}{\tau} (1 - z^{-1})$$

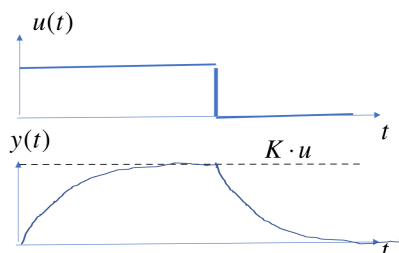
$$G(s) = \frac{K}{1 + Ts}$$



$$G(s) = \frac{Y(s)}{U(s)} \Rightarrow H(z^{-1}) = \frac{Y(z^{-1})}{U(z^{-1})}$$

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$$s = \frac{1}{\tau} (1 - z^{-1})$$

$$G(s) = \frac{K}{1 + Ts} \Rightarrow H(z^{-1}) = \frac{K}{1 + \frac{T}{\tau} (1 - z^{-1})} = \frac{Y(z^{-1})}{U(z^{-1})}$$

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$$Y \cdot \left(1 + \frac{T}{\tau} (1 - z^{-1}) \right) = K \cdot U$$

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$$(\tau + T) \cdot Y + T \cdot Y \cdot z^{-1} = \tau \cdot K \cdot U$$

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$$(\tau + T) \cdot y(n) + T \cdot y(n-1) = \tau \cdot K \cdot u(n)$$

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$$(\tau + T) \cdot y(n) + T \cdot y(n - 1) = \tau \cdot K \cdot u(n)$$

$$y(n) = \frac{T}{(\tau + T)} y(n-1) + \frac{\tau}{(\tau + T)} \cdot K \cdot u(n)$$

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$\tau \ll T$ $\tau = 0.2 \text{ s}, \quad T = 0.8 \text{ s}, \quad K = 2$

$$\tau \leq \frac{1}{5}T \quad y(n) = 0.8 \cdot y(n-1) + 0.2 \cdot 2 \cdot u(n)$$

[illegible]

$$y(n) = \frac{T}{(\tau + T)} y(n-1) + \frac{\tau}{(\tau + T)} \cdot K \cdot u(n)$$

$\tau \ll T$
 $\tau \leq \frac{1}{5} T$

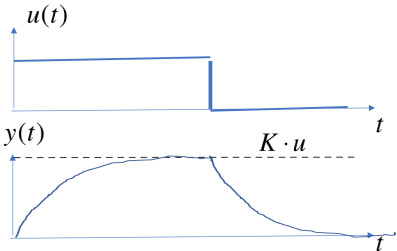
 $\tau = 0.2 \text{ s}, \quad T = 0.8 \text{ s}, \quad K = 2$
 $y(n) = 0.8 \cdot y(n-1) + 0.2 \cdot 2 \cdot u(n)$

n	1	2	3	4	5	6	...	N	N+1	N+2	N+3	...	M	M+1
u(n)	1	1	1	1	1	1	1	1	1	0	0	0	0	0
y(n)	0	0.4	0.72	0.98	1.18	1.34		2.00	2.00	1.6	1.28		0	0

$$y(n) = \frac{T}{(\tau + T)} y(n-1) + \frac{\tau}{(\tau + T)} \cdot K \cdot u(n)$$

n	1	2	3	4	5	6	...	N	N+1	N+2	N+3	...	M	M+1
u(n)	1	1	1	1	1	1	1	1	1	0	0	0	0	0
y(n)	0	0.4	0.72	0.98	1.18	1.34		2.00	2.00	1.6	1.28		0	0

$$G(s) = \frac{K}{1 + Ts}$$



- ✓ Understand the concepts
- ✓ Understand the math
- ✓ Work examples by hand
- Solve problems on computer